ACO-Based Edge Detection

Ant Colony Optimization for Image Edge Detection

Mini Project: Image Analysis & Computer Vision

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Type: Solo Project

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Problem Statement

Edge detection is a fundamental operation in image analysis: identifying boundaries where pixel intensity changes sharply.

Challenges with Traditional Methods:

- Fixed-kernel methods (Sobel, Canny) cannot adaptively respond to varying edge strengths
- Pixel-wise operators produce fragmented, disconnected edge contours
- Sensitive to noise without careful parameter tuning

Our Approach:

- Apply Ant Colony Optimization (ACO): a swarm intelligence metaheuristic
- Artificial ants traverse the pixel grid, depositing pheromone on edge locations
- Pheromone accumulates along true edge contours over multiple iterations
- Evaluate against human-annotated ground truth from BSDS300 (300 images)

Literature Survey

[1] Tian, Yu & Xie (2008): IEEE CEC

- Pioneered ACO for image edge detection. Ants traverse pixel grids using pheromone and gradient heuristics. Competitive with Sobel and Canny.

[2] Nezamabadi-pour et al.: MRF-Based Image Segmentation

- Extended ACO to MRF-based segmentation, showing pheromone-guided traversal captures spatial context beyond local gradient information.

[3] IJETT V71I9P233: Edge Detection Using ACO

- Comprehensive survey comparing ACO with classical methods. Analyzes parameter sensitivity. ACO produces smoother, more connected edges.

Biological Inspiration

Real Ant Behavior

- Ants deposit pheromone on paths while foraging
- Shorter, better paths accumulate more pheromone
- Other ants prefer pheromone-rich paths
- Old pheromone evaporates over time
- Collective behavior -> emergent optimization

Our Analogy

- Image pixels = graph nodes
- 8-connected neighbors = possible moves
- Gradient magnitude = heuristic desirability
- Pheromone matrix = collective edge memory
- Edge contours emerge from pheromone accumulation

Mathematical Formulation

Transition Probability

Each ant at pixel (i,j) chooses its next position from the 8-connected neighborhood:

$$P(i,j) = \frac{\tau(i,j)^{\alpha} \cdot \eta(i,j)^{\beta}}{\sum_{(m,n)\in\mathcal{N}} \tau(m,n)^{\alpha} \cdot \eta(m,n)^{\beta}}$$

Pheromone Update

$$\tau(i,j) \leftarrow (1-\rho) \cdot \tau(i,j) + \Delta\tau(i,j)$$

Where:

- t(i,j) = pheromone level n(i,j) = heuristic (Sobel gradient)
- a = pheromone weight b = heuristic weight rho = evaporation rate

System Architecture & Pipeline

The following represents the step-by-step pipeline from input image to the final edge map.

Step 1	>	Step 2	>	Step 3	>	Step 4	>	Step 5	>	Step 6
Input Image (Grayscale)		Sobel Gradient n(i,j)		Init Pheromone t = 0.1		Ant Traversal K ants, S steps		Pheromone Update		Edge Map [0, 255]

How Ants Find Edges: Pseudocode

```
1. Prepare a blank map for pheromones (trails) and a gradient map of the main image.
2. For multiple turns (iterations):
3.     Place a large number of ants randomly on the image grid.
4.     For a set number of steps:
5.         Each ant looks at its 8 neighboring pixels.
6.         It calculates the chance to move across based on:
7.             - How strong the image gradient is (heuristic)
8.             - How much pheromone other ants left (trail)
9.         The ant moves and leaves a small pheromone trace on its path.
10.    After all ants finish moving, evaporate some old pheromones so bad paths fade.
11.    Add the new pheromones from this turn to the global map.
12. Once all turns are over, the areas with the most pheromone represent the true edges.
13. Thicken the edges for visibility and return the final image.
```

Dataset: BSDS300

Berkeley Segmentation Dataset: 300 natural images with human-annotated ground truth

300	200	100	5-7
Natural Images	Training Set	Test Set	Annotators/Image

Dataset Features

- Diverse scenes: animals, landscapes, people, objects, urban
- Multiple independent human segmentations per image
- Standard benchmark in edge detection and segmentation literature

Our Processing

- All images converted to grayscale
- .seg files parsed to binary edge maps via Sobel
- 3-panel dashboard visualization per image

Hyperparameters

Parameter	Symbol	Default	Batch	Description
`num_ants`	K	2,048	8,192	Ants per iteration
`num_iterations`	T	20	50	ACO iterations
`alpha`	a	1.0	1.0	Pheromone influence weight
`beta`	b	2.0	2.0	Heuristic (gradient) weight
`decay_rate`	rho	0.1	0.1	Pheromone evaporation rate
`steps_per_ant`	S	10	10	Steps each ant takes

Key insight: b > a favors gradient-guided exploration, producing edges aligned with true intensity transitions. Batch uses 8192 ants and 50 iterations for production quality.

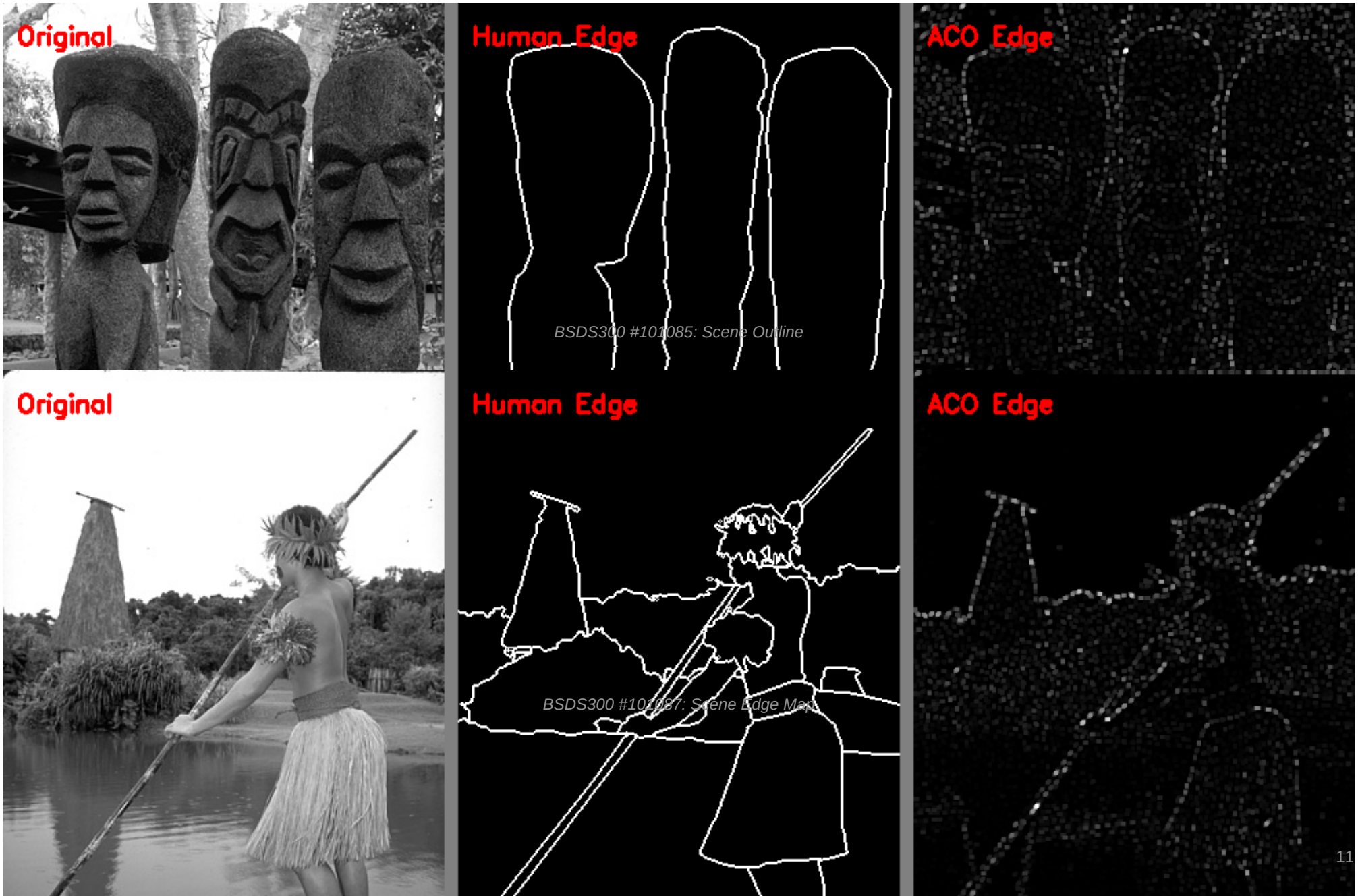
Results: Sample Outputs (Set 1)

Each dashboard: Original | Human Ground Truth | ACO Edge Map

BSDS300 #100075: Bears

BSDS300 #100098: Bear in snow

Results: Sample Outputs (Set 2)



Strengths & Limitations

Strengths

- Adaptive: pheromone reinforcement concentrates on true edges
- Connected edges: ant traversal produces continuous contours
- Noise tolerance: collective behavior averages out noise
- Flexible: tunable via a, b, rho, ant count parameters
- No training required: pure optimization approach

Limitations

- Computational cost: per-ant loops are expensive
- Parameter sensitivity: poor a/b produce weak edges
- Thinner edges: ACO edges sparser than ground truth
- No multi-scale: single-resolution Sobel heuristic
- Sequential bottleneck: hard to parallelize on CPU

ACO vs. Traditional Methods

Property	Sobel	Canny	ACO (Ours)
Approach	Fixed 3x3 kernels	Multi-stage pipeline	Swarm intelligence
Adaptivity	None	Threshold-based	Pheromone-guided
Edge Continuity	Low (pixel-wise)	Medium (hysteresis)	High (ant paths)
Noise Handling	Poor	Good (Gaussian blur)	Good (collective avg.)
Speed	Very fast	Fast	Slow (iterative)
Parameters	Kernel size	2 thresholds + sigma	a, b, rho, K, T, S

Key takeaway: ACO trades computational speed for adaptivity and edge continuity. Best suited for applications where edge quality matters more than real-time performance.

Conclusion

Final Remarks

- Successfully implemented an Ant Colony Optimization algorithm for edge detection.
- Evaluated thoroughly on 300 images from the Berkeley Segmentation Dataset (BSDS300).
- Demonstrated the ability of artificial ants to form coherent, continuous edge contours.
- Proved the algorithm works in a fully unsupervised manner without external training.
- Created an automated dashboard to visually compare outputs with human ground truth.